

DDPM Training Loss Derivation

Step 1: Log-Likelihood Objective

Maximize $\log p_{\theta}(x_0)$



Step 2: ELBO Lower Bound

$$\log p_{\theta}(x_0) \geq \text{ELBO}(\theta) = \mathbb{E}_q \left[\log \frac{p_{\theta}(x_{0:T})}{q(x_{1:T} | x_0)} \right]$$



Step 3: Decompose ELBO into KL Terms

Reconstruction : $\mathbb{E}_q [\log p_{\theta}(x_0 | x_1)]$

Prior matching : $-D_{\text{KL}}(q(x_T | x_0) \parallel p(x_T))$

Denoising matching : $-\sum_{t=2}^T D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \parallel p_{\theta}(x_{t-1} | x_t))$



Step 4: Posterior Is Gaussian

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \sigma_q^2 I)$$



Step 5: KL Between Gaussians $\rightarrow$ Mean Matching

$$D_{\text{KL}}(q \parallel p_{\theta}) = \frac{1}{2\sigma_q^2} \|\mu_q(x_t, x_0) - \mu_{\theta}(x_t, t)\|^2 + \text{const}$$



Step 6: Reparameterize x_0 in Terms of Noise

$$x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}} (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon), \quad \epsilon \sim \mathcal{N}(0, I)$$



Step 7: Predict Noise via ϵ_{θ}

$$\mu_{\theta}(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(x_t, t) \right)$$



Simplified Training Loss

$$\mathcal{L}_{\text{simple}} = \mathbb{E}_{t, x_0, \epsilon} \left[\left\| \epsilon - \epsilon_{\theta}(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t) \right\|^2 \right]$$

Predict the noise ϵ that was added at each step t .